Math 54 — Week 5

Properties of determinants; Cramer's rule and volume; vector spaces; null and column spaces (Lay 3.2–3.3, 4.1–4.2)

§1 Facts used from Weeks 1–4

(W1) (Week 4) $\det A$ can be computed by cofactor expansion along any row or down any column; for a triangular matrix it is the product of the diagonal entries.

(W2) (Week 4) An elementary matrix E performs a row operation: EA is A after that operation. The inverse of an elementary matrix is elementary. If A is invertible, then $E_k \cdots E_1 A = I$ for some elementary E_i, so $A = E_1^{-1} \cdots E_k^{-1}$ is a product of elementary matrices.

(W3) (Week 4) Invertible matrix theorem: for square A, invertible $\iff$ n pivots $\iff$ $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\iff$ the columns are independent $\iff$ the columns span $\mathbb{R}^n$ $\iff$ A^T is invertible. If A, B are square and AB is invertible, so are A and B. Column j of A^{-1} solves $A\mathbf{x} = \mathbf{e}_j$. $(AB)^T = B^T A^T$.

(W4) (Week 2) The solution set of $A\mathbf{x} = \mathbf{0}$ is the span of the vectors in its parametric vector form. $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $\mathbf{b}$ is a combination of the columns of A, and this holds for every $\mathbf{b}$ exactly when A has a pivot in every row.

(W5) (Weeks 2–3) In an echelon form, each nonzero row has a pivot and the zero rows are at the bottom. In an $n \times n$ echelon form with n pivots, the pivot of row i is in column i (the argument of Week 4 for the reduced form uses only the echelon conditions).

§2 This week

§2.1 Properties of determinants (Lay 3.2)

The question. Expanding an $n \times n$ determinant by cofactors needs n determinants of size $n-1$, so at least $n!$ multiplications: $10! = 3\,628\,800$ for a 10×10 matrix. Row reduction of a 10×10 matrix to echelon form takes about $n^3/3 \approx 330$ multiplications: clearing the entries below pivot k takes about $(n-k)^2$ of them, and $\sum_k (n-k)^2 \approx n^3/3$. How does each row operation change the determinant, so that row reduction can compute it? For example, what is $\det A$ for the matrix below?

$$A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$$

Theorem (Row operations (Lay 3.2, Thm. 3))

Let A be a square matrix.

(a) If a multiple of one row of A is added to another row to produce B, then $\det B = \det A$.

(b) If two rows of A are interchanged to produce B, then $\det B = -\det A$.

(c) If one row of A is multiplied by k to produce B, then $\det B = k \det A$.

The idea

A cofactor expansion along a row that the operation did not change uses the same entries and signs for A and B, and minors that differ by the same kind of operation one size smaller. So each statement for size n follows from the statement for size $n-1$, starting from a direct 2×2 computation.

Proof. 1. (c) Expand $\det B$ along the scaled row i. Its entries are ka_{ij}, and its cofactors are those of A, because deleting row i removes the only change. So $\det B = \sum_j ka_{ij}C_{ij} = k \det A$.

2. (a), (b) for $n = 2$. Interchanging the rows, and adding k times row 1 to row 2:

$$\det \begin{bmatrix} c & d \\ a & b \end{bmatrix} = cb - da = -(ad - bc), \qquad \det \begin{bmatrix} a & b \\ c + ka & d + kb \end{bmatrix} = a(d + kb) - b(c + ka) = ad - bc.$$

Adding k times row 2 to row 1:

$$\det \begin{bmatrix} a + kc & b + kd \\ c & d \end{bmatrix} = (a + kc)d - (b + kd)c = ad - bc.$$

3. (a), (b) for $n \geq 3$, **assuming them for size** $n - 1$. The operation involves at most two rows, so some row i is unchanged. Expanding along row i,

$$\det B = \sum_j a_{ij}(-1)^{i+j} \det B_{ij},$$

and each B_{ij} is A_{ij} after the same kind of operation. By the case $n - 1$, $\det B_{ij} = \det A_{ij}$ for a replacement and $\det B_{ij} = -\det A_{ij}$ for an interchange. Summing over j gives $\det B = \det A$, respectively $\det B = -\det A$.

□

Determinant by row reduction. Reduce A to an echelon form U using only replacements and r interchanges. Each replacement keeps the determinant and each interchange changes its sign, so $\det A = (-1)^r \det U$; U is square and in echelon form, hence triangular, so by (W1)

$$\det A = (-1)^r \cdot (\text{product of the diagonal entries of } U).$$

The question answered.

$$\begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix} \xrightarrow[R_3 + R_1]{R_2 + 2R_1} \begin{bmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{bmatrix}.$$

One interchange, so $\det A = (-1)^1 \cdot (1)(3)(-5) = 15$.

Two equal rows, or a zero row, give determinant 0. If rows i and j are equal, the replacement $R_i - R_j$ keeps the determinant and produces a zero row. Expanding along a zero row gives 0.

Theorem (Invertibility (Lay 3.2, Thm. 4))

A square matrix A is invertible if and only if $\det A \neq 0$.

Proof. 1. With U as above, $\det A = \pm \det U$, so $\det A \neq 0$ exactly when $\det U \neq 0$.

2. **Assume** A **has** n **pivots.** By (W5) the pivots of U are its diagonal entries, so $\det U$ is the product of the pivots, which is not 0.

3. **Assume A has fewer than n pivots.** Then U has a row without a pivot, which is a zero row and lies at the bottom (W5). So the last diagonal entry of U is 0 and $\det U = 0$.
4. So $\det A \neq 0$ exactly when A has n pivots, which by the invertible matrix theorem (W3) is exactly when A is invertible.

□

This adds a statement to the invertible matrix theorem. In particular, **the columns of a square matrix are linearly independent if and only if** $\det A \neq 0$.

Theorem (Transpose (Lay 3.2, Thm. 5))
If A is $n \times n$, then $\det A^T = \det A$.

Proof. 1. By induction on n. For $n = 1$, $A^T = A$.

2. **Assume the statement for size $n-1$.** Deleting row 1 and column j of A^T gives $(A_{j1})^T$, so the expansion of $\det A^T$ along row 1 is $\sum_j a_{j1}(-1)^{1+j} \det(A_{j1})^T$.
3. By the assumption, $\det(A_{j1})^T = \det A_{j1}$, so this sum is the expansion of $\det A$ down column 1, which equals $\det A$ (W1).

□

Rows of A^T are columns of A, so **column operations change the determinant exactly as row operations do**: a column replacement keeps it, a column interchange changes its sign, and scaling a column by k multiplies it by k.

Theorem (Multiplicative property (Lay 3.2, Thm. 6))
If A and B are $n \times n$, then $\det(AB) = (\det A)(\det B)$.

Proof. 1. **For an elementary E and any M, $\det(EM) = (\det E)(\det M)$.** EM is M after one row operation (W2), so by Theorem 3 $\det(EM)$ is $\det M$, $-\det M$ or $k \det M$. The same theorem applied to $M = I$ gives $\det E = 1$, -1 or k for the same operation.

2. **Assume A is not invertible.** Then AB is not invertible (W3), so by Theorem 4 both $\det(AB)$ and $(\det A)(\det B)$ are 0.
3. **Assume A is invertible.** Then $A = E_p \cdots E_1$ for elementary matrices (W2). Applying step 1 p times,

$$\det(AB) = \det(E_p \cdots E_1 B) = (\det E_p) \cdots (\det E_1) \det B, \qquad \det A = (\det E_p) \cdots (\det E_1).$$

So $\det(AB) = (\det A)(\det B)$.

□

Consequences. For $n \times n$ matrices:

- If A is invertible, $(\det A)(\det A^{-1}) = \det(AA^{-1}) = \det I = 1$, so

$$\det(A^{-1}) = \frac{1}{\det A}.$$

- $\det(A^k) = (\det A)^k$, by Theorem 6 applied $k-1$ times.
- $\det(cA) = c^n \det A$, because $cA = (cI)A$ and $\det(cI) = c^n$ (cI is diagonal).
- **Linearity in one column**: with all other columns fixed, $\mathbf{x} \mapsto \det[\,\mathbf{a}_1 \cdots \mathbf{x} \cdots \mathbf{a}_n\,]$ is a linear function of $\mathbf{x}$. Reason: expanding down that column gives $x_1 C_{1j} + \cdots + x_n C_{nj}$, and the cofactors C_{ij} do not involve $\mathbf{x}$.

Shortcuts

- **Determinant by row reduction**: use only replacements and interchanges, count the interchanges, multiply the diagonal. Reason: Theorem 3. Instance: the question.

- **Pull out a common factor** of a row (or column) first; it multiplies the determinant by that factor. Reason: Theorem 3(c). Instance: row 1 of P1 is $2(1, 2, -1, 3)$, so det of P1's matrix is 2 times the determinant with row 1 replaced by $(1, 2, -1, 3)$.
- **Combine expansion and reduction**: create zeros in one column with replacements (which keep det), then expand down that column. Reason: Theorem 3(a) and (W1). Instance: $R_2 - 2R_1$ and $R_3 - 3R_1$ give

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 5 & 9 \end{bmatrix} = \det \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} = 1 \cdot \det \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 1,$$

  expanding down column 1.
- **Invertible for which h?** Compute det as a polynomial in h; the matrix is invertible exactly away from its zeros. Reason: Theorem 4. Instance: the matrix below has $\det = 4 - 2h$, so it is invertible exactly when $h \neq 2$.

$$\begin{bmatrix} 1 & h \\ 2 & 4 \end{bmatrix}$$

- **Determinant rules**: $\det(AB) = \det A \det B$, $\det A^{-1} = 1/\det A$, $\det A^T = \det A$, $\det(cA) = c^n \det A$. Instance: for 3×3 A with $\det A = 5$, $\det(2A^{-1}) = 2^3 \cdot \frac{1}{5} = \frac{8}{5}$.

Traps

- Scaling a row by k during row reduction multiplies det by k; divide it back out. Scaling row 1 of the matrix below on the left by $\frac{1}{2}$ gives the matrix on the right, with determinant 3, while the original determinant is $2 \cdot 3 = 6$.

$$\begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$$

- $\det(cA) = c^n \det A$, not $c \det A$: $\det(2I_2) = 4$. And $\det(A + B) \neq \det A + \det B$ in general: for 2×2, $\det(I + (-I)) = 0$ but $\det I + \det(-I) = 2$.
- $\det A = 0$ does not require two equal rows or a zero row. The matrix below has neither, and no two of its rows are multiples of each other, yet row 3 = 2(row 2) − row 1, so det = 0.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

- "Product of the diagonal" is the determinant only for triangular matrices: the matrix below has diagonal product 0 and determinant -1.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

§2.2 Cramer's rule, the inverse formula, and volume (Lay 3.3)

The question. Can the system $2x_1 + x_2 = 7$, $-3x_1 + 4x_2 = 6$ be solved with determinants alone, without row reduction? And what is the area of the parallelogram with sides $(2, -3)$ and $(1, 4)$?

Notation. For an $n \times n$ matrix A and $\mathbf{b}$ in $\mathbb{R}^n$, $A_i(\mathbf{b})$ is A with column i replaced by $\mathbf{b}$.

Theorem (Cramer's rule (Lay 3.3, Thm. 7))

If A is an invertible $n \times n$ matrix, then for each $\mathbf{b}$ the unique solution of $A\mathbf{x} = \mathbf{b}$ has entries

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \qquad i = 1, \dots, n.$$

The idea

The identity matrix with column i replaced by $\mathbf{x}$ has determinant x_i, and multiplying it by A gives A with column i replaced by $A\mathbf{x} = \mathbf{b}$. Taking determinants of that product gives the formula.

Proof. 1. Let $I_i(\mathbf{x})$ be I with column i replaced by the solution $\mathbf{x}$. Column $j \neq i$ of $AI_i(\mathbf{x})$ is $A\mathbf{e}_j = \mathbf{a}_j$, and column i is $A\mathbf{x} = \mathbf{b}$, so $AI_i(\mathbf{x}) = A_i(\mathbf{b})$.

2. Expand $\det I_i(\mathbf{x})$ along row i: its only nonzero entry is x_i in position (i, i), with sign $+$, and deleting row i and column i leaves I_{n-1}. So $\det I_i(\mathbf{x}) = x_i$.

3. By Theorem 6, $(\det A)\, x_i = \det A_i(\mathbf{b})$. Divide by $\det A \neq 0$ (Theorem 4).

□

The question answered (system).

$$\det A = \det \begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix} = 11, \quad \det A_1(\mathbf{b}) = \det \begin{bmatrix} 7 & 1 \\ 6 & 4 \end{bmatrix} = 22, \quad \det A_2(\mathbf{b}) = \det \begin{bmatrix} 2 & 7 \\ -3 & 6 \end{bmatrix} = 33,$$

so $x_1 = 22/11 = 2$ and $x_2 = 33/11 = 3$. Check: $4 + 3 = 7$ and $-6 + 12 = 6$.

Worked example: Cramer's rule with a parameter. For which s does $sx_1 + 2x_2 = 1$, $3x_1 + sx_2 = 2$ have a unique solution, and what is it?

1. $\det A = s^2 - 6$, so the solution is unique exactly when $s \neq \pm\sqrt{6}$ (Theorem 4).

2. For those s,

$$x_1 = \frac{1}{s^2 - 6} \det \begin{bmatrix} 1 & 2 \\ 2 & s \end{bmatrix} = \frac{s - 4}{s^2 - 6}, \qquad x_2 = \frac{1}{s^2 - 6} \det \begin{bmatrix} s & 1 \\ 3 & 2 \end{bmatrix} = \frac{2s - 3}{s^2 - 6}.$$

3. Check with $s = 0$: the system is $2x_2 = 1$, $3x_1 = 2$, so $x_1 = \frac{2}{3}$ and $x_2 = \frac{1}{2}$; the formulas give $\frac{-4}{-6}$ and $\frac{-3}{-6}$.

Theorem (An inverse formula (Lay 3.3, Thm. 8))

If A is invertible, then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A,$$

where the **adjugate** $\operatorname{adj} A$ is the *transpose* of the matrix of cofactors: $(\operatorname{adj} A)_{ij} = C_{ji}$.

Proof. 1. Column j of A^{-1} is the solution of $A\mathbf{x} = \mathbf{e}_j$ (W3). By Cramer's rule, its entry i is $\det A_i(\mathbf{e}_j)/\det A$.

2. Column i of $A_i(\mathbf{e}_j)$ is $\mathbf{e}_j$, with a single 1 in row j. Expanding down column i gives $(-1)^{j+i}$ times the determinant of $A_i(\mathbf{e}_j)$ without row j and column i, which is A without row j and column i. So $\det A_i(\mathbf{e}_j) = (-1)^{i+j} \det A_{ji} = C_{ji}$.

□

Worked example: a 3×3 adjugate. For the matrix below, expansion along row 1 gives $\det A = 1 \cdot 1 - 2 \cdot (0 - 1) + 0 = 3$. The nine cofactors are:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \qquad \begin{aligned} C_{11} &= +\det \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = 1, & C_{12} &= -\det \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = 1, & C_{13} &= +\det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1, \\ C_{21} &= -\det \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = -2, & C_{22} &= +\det \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = 1, & C_{23} &= -\det \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} = 2, \\ C_{31} &= +\det \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = 2, & C_{32} &= -\det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -1, & C_{33} &= +\det \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = 1. \end{aligned}$$

The adjugate has the cofactors C_{1j} as its *column* 1, and so on:

$$\operatorname{adj} A = \begin{bmatrix} 1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{bmatrix}, \qquad A^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{bmatrix}.$$

Check one column: row 1 of A times column 1 of $\operatorname{adj} A$ is $1 + 2 + 0 = 3$, and row 2 times column 1 is $0 + 1 - 1 = 0$, as $A \operatorname{adj} A = 3I$ requires. For numerical work, row reducing $[\,A \;\; I\,]$ is faster; the formula is for 2×2 matrices (it gives the formula of Week 4) and for theory.

Determinants as area and volume.

Theorem (Area and volume (Lay 3.3, Thm. 9))

If A is 2×2, the area of the parallelogram determined by the columns of A is $|\det A|$. If A is 3×3, the volume of the parallelepiped determined by the columns of A is $|\det A|$.

The idea

A column replacement $\mathbf{a}_2 \to \mathbf{a}_2 + c\mathbf{a}_1$ slides the far side of the parallelogram along the line of $\mathbf{a}_1$: the base and the height stay the same, so the area does, and the determinant does too. Such moves turn any parallelogram into a rectangle with sides along the axes, where area and $|\det|$ are both the product of the side lengths.

Proof for 2×2. 1. **Assume the columns are dependent.** Then they lie on one line through $\mathbf{0}$, the parallelogram is flat with area 0, and $\det A = 0$ (Theorem 4).

2. **Assume the columns are independent**, so $\det A \neq 0$. A column replacement $\mathbf{a}_2 \to \mathbf{a}_2 + c\mathbf{a}_1$ keeps det (column version of Theorem 3(a)) and keeps the area: the new parallelogram has the same base $\mathbf{a}_1$, and $\mathbf{a}_2 + c\mathbf{a}_1$ differs from $\mathbf{a}_2$ by a vector along $\mathbf{a}_1$, so it is the same distance from the line of $\mathbf{a}_1$ (same height). The same holds for $\mathbf{a}_1 \to \mathbf{a}_1 + c\mathbf{a}_2$. A column interchange keeps $|\det|$ and the parallelogram.

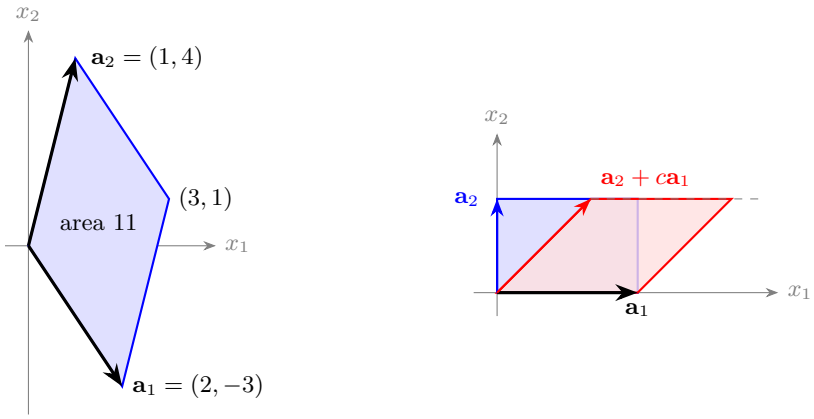
3. Column operations on A are row operations on A^T. Reduce A^T to echelon form by replacements and interchanges; it has two pivots $p, r \neq 0$, so it is upper triangular, and A has become its transpose, which is lower triangular:

$$A^T \to \begin{bmatrix} p & q \\ 0 & r \end{bmatrix}, \qquad A \to \begin{bmatrix} p & 0 \\ q & r \end{bmatrix}.$$

   One more column replacement, column 1 minus $\frac{q}{r}$ times column 2, gives $\operatorname{diag}(p, r)$.

4. The parallelogram of $\operatorname{diag}(p, r)$ is a rectangle with sides $|p|$ and $|r|$, of area $|pr| = |\det \operatorname{diag}(p, r)|$. All the steps kept both the area and $|\det|$, so the area of the original parallelogram is $|\det A|$.

The 3×3 case is proved the same way with parallelepipeds (base area times height); it is cited here. □



Left: the question's parallelogram, with sides $\mathbf{a}_1$, $\mathbf{a}_2$ and fourth vertex $\mathbf{a}_1 + \mathbf{a}_2 = (3,1)$. Right: a column replacement keeps the base $\mathbf{a}_1$ and the height (the top side stays on the dashed line), so it keeps the area.

The question answered (area). The parallelogram has area $|\det A| = |2 \cdot 4 - 1 \cdot (-3)| = 11$.

Parallelograms not at the origin. The parallelogram with vertices $\mathbf{p}$, $\mathbf{p}+\mathbf{u}$, $\mathbf{p}+\mathbf{v}$, $\mathbf{p}+\mathbf{u}+\mathbf{v}$ is the one with sides $\mathbf{u}, \mathbf{v}$ moved by $\mathbf{p}$. Moving a region without turning it does not change its area, so its area is $|\det[\,\mathbf{u}\ \ \mathbf{v}\,]|$.

Theorem (Linear maps scale area and volume (Lay 3.3, Thm. 10))
Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be $T(\mathbf{x}) = A\mathbf{x}$. If S is a parallelogram, then $\text{area}(T(S)) = |\det A| \cdot \text{area}(S)$. The same holds for every region S with finite area, and in $\mathbb{R}^3$ with volume in place of area.

Proof for parallelograms. 1. Let S have sides $\mathbf{b}_1, \mathbf{b}_2$ at $\mathbf{0}$: $S = \{s_1\mathbf{b}_1 + s_2\mathbf{b}_2 : 0 \le s_1, s_2 \le 1\}$. By linearity, $T(S) = \{s_1 A\mathbf{b}_1 + s_2 A\mathbf{b}_2\}$, the parallelogram with sides $A\mathbf{b}_1$ and $A\mathbf{b}_2$.

2. With $B = [\,\mathbf{b}_1\ \ \mathbf{b}_2\,]$, Theorem 9 and Theorem 6 give

$$\text{area}(T(S)) = |\det[\,A\mathbf{b}_1\ \ A\mathbf{b}_2\,]| = |\det(AB)| = |\det A|\,|\det B| = |\det A| \cdot \text{area}(S).$$

3. A parallelogram $\mathbf{p} + S$ goes to $T(\mathbf{p}) + T(S)$, a moved copy of $T(S)$, with the same area.

For general regions, Lay approximates the region by small squares; that step is cited here. □

Worked example: area of an ellipse. The ellipse $x_1^2/a^2 + x_2^2/b^2 \le 1$ $(a, b > 0)$ is the image of the unit disk under $\mathbf{x} \mapsto \text{diag}(a,b)\mathbf{x}$, because (u_1, u_2) with $u_1^2 + u_2^2 \le 1$ goes to (au_1, bu_2). So its area is $|ab| \cdot \pi = \pi ab$.

Shortcuts

- **Cramer's rule** is fastest for a 2×2 system, or for one entry x_i of a larger system. Reason: each entry costs one extra determinant. Instance: the question.
- **Area of a triangle** with vertices $\mathbf{0}, \mathbf{u}, \mathbf{v}$: half the parallelogram, $\frac{1}{2}|\det[\,\mathbf{u}\ \mathbf{v}\,]|$. Instance: $\mathbf{0}$, $(3,0)$, $(1,2)$ give $\frac{1}{2}|6 - 0| = 3$.
- **Triangle with vertices** (x_i, y_i): the area is half the absolute value of

$$\det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$$

  Reason: $R_2 - R_1$ and $R_3 - R_1$ keep the determinant and make column 3 equal to $(1,0,0)$; expanding down column 3 leaves the 2×2 determinant below, which by Theorem 5 is $\det[\,\mathbf{u}\ \mathbf{v}\,]$ for the sides $\mathbf{u}, \mathbf{v}$ from vertex 1.

$$\det \begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{bmatrix}$$

  Instance: $(1,1)$, $(4,1)$, $(2,3)$ give $\frac{1}{2}|3 \cdot 2 - 0 \cdot 1| = 3$.

- **Area of an image**: multiply the area by $|\det A|$. Reason: Theorem 10. Instance: the ellipse.

Traps

- The adjugate is the *transpose* of the cofactor matrix. For the matrix below, the cofactors are $C_{11} = 4$, $C_{12} = -3$, $C_{21} = -2$, $C_{22} = 1$, so the adjugate is the transposed arrangement:

$$\operatorname{adj}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} = \begin{bmatrix}4 & -2\\ -3 & 1\end{bmatrix}, \qquad \text{not} \qquad \begin{bmatrix}4 & -3\\ -2 & 1\end{bmatrix}.$$

- Area is $|\det|$; the sign of det records orientation. The columns $(0,1)$ and $(1,0)$ give det $= -1$ and area 1.
- Cramer's rule needs $\det A \neq 0$. In the parameter example with $s = \sqrt{6}$ the formulas divide by 0; then the system has no solution or infinitely many.

§2.3 Vector spaces and subspaces (Lay 4.1)

The question. Polynomials, functions and matrices can all be added and scaled like vectors in $\mathbb{R}^n$. Among the polynomials of degree at most 2, do those with $p(1) = 0$ form a "space" in which sums and multiples stay inside, like a plane through the origin? Do those with $p(1) = 1$? What exactly is needed for span and linear combination to work, and how do we test a set?

Definition. A **vector space** is a nonempty set V of objects (**vectors**) with an addition and a multiplication by real scalars such that, for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V and all scalars c, d:

(1) $\mathbf{u} + \mathbf{v}$ is in V;
(2) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
(3) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;
(4) there is a $\mathbf{0}$ in V with $\mathbf{u} + \mathbf{0} = \mathbf{u}$;
(5) each $\mathbf{u}$ has a $-\mathbf{u}$ in V with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$;
(6) $c\mathbf{u}$ is in V;
(7) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$;
(8) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$;
(9) $c(d\mathbf{u}) = (cd)\mathbf{u}$;
(10) $1\mathbf{u} = \mathbf{u}$.

Examples. In each, the operations are the familiar ones and the axioms reduce to rules for numbers.

- $\mathbb{R}^n$, with zero vector $(0, \ldots, 0)$.
- $\mathbb{P}_n$, the polynomials $a_0 + a_1 t + \cdots + a_n t^n$ of degree at most n, added and scaled coefficient by coefficient; the zero vector is the zero polynomial, all of whose coefficients are 0. $\mathbb{P}$ is the space of all polynomials.
- The real-valued functions on a set D, with $(f + g)(t) = f(t) + g(t)$ and $(cf)(t) = cf(t)$; the zero vector is the zero function, with value 0 at every t. The continuous functions on $[a, b]$ form the space $C[a, b]$.
- The $m \times n$ matrices, with the zero matrix as zero vector.

Theorem (Consequences of the axioms (Lay 4.1))
In every vector space, for every $\mathbf{u}$ and scalar c: $0\mathbf{u} = \mathbf{0}$, $c\mathbf{0} = \mathbf{0}$, and $-\mathbf{u} = (-1)\mathbf{u}$.

Proof. 1. By (8), $0\mathbf{u} = (0 + 0)\mathbf{u} = 0\mathbf{u} + 0\mathbf{u}$. Add $-(0\mathbf{u})$ to both sides and use (3), (5), (4): $\mathbf{0} = 0\mathbf{u}$.

2. By (4) and (7), $c\mathbf{0} = c(\mathbf{0} + \mathbf{0}) = c\mathbf{0} + c\mathbf{0}$. Add $-(c\mathbf{0})$ to both sides as in step 1: $\mathbf{0} = c\mathbf{0}$.

3. A vector has only one negative: if $\mathbf{u} + \mathbf{w} = \mathbf{0}$ and $\mathbf{u} + \mathbf{w}' = \mathbf{0}$, then $\mathbf{w} = \mathbf{w} + (\mathbf{u} + \mathbf{w}') = (\mathbf{w} + \mathbf{u}) + \mathbf{w}' = \mathbf{w}'$, by (4), (3), (2).

4. By (10), (8) and step 1, $\mathbf{u} + (-1)\mathbf{u} = 1\mathbf{u} + (-1)\mathbf{u} = (1 + (-1))\mathbf{u} = 0\mathbf{u} = \mathbf{0}$. So $(-1)\mathbf{u}$ is the negative of $\mathbf{u}$.

□

Definition. A **subspace** of a vector space V is a subset H of V with three properties: (a) $\mathbf{0}$ is in H; (b) H is closed under addition: $\mathbf{u}+\mathbf{v}$ is in H whenever $\mathbf{u},\mathbf{v}$ are; (c) H is closed under scalar multiplication: $c\mathbf{u}$ is in H whenever $\mathbf{u}$ is in H and c is a scalar.

A subspace is a vector space with the operations of V: axioms (1), (6) are (b), (c); (4) holds by (a); (5) holds because $-\mathbf{u}=(-1)\mathbf{u}$ is in H by (c) and the theorem above; the other axioms are equations that hold for all vectors of V, so in particular for those of H.

- $\{\mathbf{0}\}$ and V are subspaces of V. $\mathbb{P}_n$ is a subspace of $\mathbb{P}$.
- A line through the origin in $\mathbb{R}^2$ is a subspace; a line not through the origin is not (it misses $\mathbf{0}$).
- $\mathbb{R}^2$ is *not* a subset, hence not a subspace, of $\mathbb{R}^3$: its vectors have two entries, not three. The plane $\{(s,t,0)\}$ in $\mathbb{R}^3$ is a subspace.

Theorem (A span is a subspace (Lay 4.1, Thm. 1))
If $\mathbf{v}_1,\dots,\mathbf{v}_p$ are in a vector space V, then $\mathrm{Span}\{\mathbf{v}_1,\dots,\mathbf{v}_p\}$ is a subspace of V.

Proof. 1. $\mathbf{0}=0\mathbf{v}_1+\cdots+0\mathbf{v}_p$ (by the theorem above), so $\mathbf{0}$ is in the span.

2. By the axioms, $\sum c_i\mathbf{v}_i+\sum d_i\mathbf{v}_i=\sum(c_i+d_i)\mathbf{v}_i$ and $k\sum c_i\mathbf{v}_i=\sum(kc_i)\mathbf{v}_i$; both are again combinations of $\mathbf{v}_1,\dots,\mathbf{v}_p$.

□

Worked example: a set given by parameters. $H=\{(s+t,\ s,\ 2t) : s,t \text{ real}\}$. Splitting by parameter, $(s+t,s,2t)=s(1,1,0)+t(1,0,2)$, so $H=\mathrm{Span}\{(1,1,0),(1,0,2)\}$, a subspace of $\mathbb{R}^3$ by Theorem 1.

The question answered. $H=\{p\in\mathbb{P}_2 : p(1)=0\}$ is a subspace of $\mathbb{P}_2$: the zero polynomial has value 0 at 1; if $p(1)=q(1)=0$ then $(p+q)(1)=p(1)+q(1)=0$; and $(cp)(1)=c\,p(1)=0$. The set $\{p\in\mathbb{P}_2 : p(1)=1\}$ is not, since the zero polynomial has value $0\neq 1$ at 1. Span and linear combination make sense in any set closed under sums and multiples, and the three subspace conditions test exactly that.

Shortcuts

- **Is H a subspace?** First check $\mathbf{0}\in H$; if not, stop. Reason: condition (a). Instance: $\{p : p(1)=1\}$.
- **"All vectors of the form ..."** with expressions linear in the parameters and no constant terms: split by parameter; it is a span, hence a subspace (Theorem 1). Instance: the worked example.
- **A condition preserved by sums and multiples**: check (b) and (c) directly, as in the question.

Traps

- The set defined by $x_1+x_2=1$ is not a subspace: it misses $\mathbf{0}$.
- Closed under addition is not enough. The polynomials with integer coefficients contain $\mathbf{0}$ and are closed under addition, but $\frac{1}{2}\cdot 1=\frac{1}{2}$ does not have an integer coefficient, so (c) fails.
- Closed under scalar multiplication is not enough either. The set $\{(a,b) : ab\geq 0\}$ (the first and third quadrants) contains every multiple of each of its vectors, but $(1,0)+(0,-1)=(1,-1)$ is not in it.
- The three conditions must hold for *all* vectors and scalars; one example of a sum that stays inside proves nothing.

§2.4 Null spaces, column spaces, and linear transformations (Lay 4.2)

The question. For the 3×5 matrix A below, is $(2, 1, 0, 0, 0)$ a solution of $A\mathbf{x} = \mathbf{0}$? Is $(1, 2, 4)$ a combination of the columns of A? Is $(1, 2, 3)$? How do we describe all such vectors?

$$A = \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 2 & -4 & 1 & 0 & 4 \end{bmatrix}$$

Definition. For an $m \times n$ matrix A: the **null space** is $\operatorname{Nul} A = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$; the **column space** $\operatorname{Col} A$ is the span of the columns of A, a subset of $\mathbb{R}^m$; the **row space** $\operatorname{Row} A$ is the span of the rows of A, written as vectors in $\mathbb{R}^n$.

The rows of A are the columns of A^T, so $\operatorname{Row} A = \operatorname{Col} A^T$. By (W4), $\operatorname{Col} A = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$, and $\operatorname{Col} A = \mathbb{R}^m$ exactly when A has a pivot in every row.

Theorem (Null space (Lay 4.2, Thm. 2) and column space (Lay 4.2, Thm. 3))
$\operatorname{Nul} A$ is a subspace of $\mathbb{R}^n$, and $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$.

Proof. 1. $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0}$ is in $\operatorname{Nul} A$.

2. If $A\mathbf{u} = \mathbf{0}$ and $A\mathbf{v} = \mathbf{0}$, then by linearity $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0}$ and $A(c\mathbf{u}) = c(A\mathbf{u}) = \mathbf{0}$.

3. $\operatorname{Col} A$ is a span, so it is a subspace by Theorem 1.

□

An explicit description of $\operatorname{Nul} A$. Solve $A\mathbf{x} = \mathbf{0}$ in parametric vector form; the vectors attached to the free variables span $\operatorname{Nul} A$ (W4). For the question's matrix, $R_3 - 2R_1$ and then $R_3 - R_2$ give the reduced echelon form:

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 2 & -4 & 1 & 0 & 4 \end{bmatrix} \xrightarrow{R_3 - 2R_1} \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 1 & 2 & -2 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

So $x_1 = 2x_2 + x_4 - 3x_5$ and $x_3 = -2x_4 + 2x_5$, with x_2, x_4, x_5 free, and

$$\operatorname{Nul} A = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

This spanning set is linearly independent. The vector for the free variable x_k has 1 in entry k and 0 in the entries of the other free variables. So in a combination $c_2\mathbf{u} + c_4\mathbf{v} + c_5\mathbf{w}$, entries $2, 4, 5$ are c_2, c_4, c_5, and the combination is $\mathbf{0}$ only if all three weights are 0.

Writing a subspace as a null space or a column space.

- **Homogeneous equations give a null space.** $\{(a, b, c) : a + b = 2c,\ b = 3a\}$ is the set of solutions of $a + b - 2c = 0$, $-3a + b = 0$, so it is Nul of the matrix below.

$$\begin{bmatrix} 1 & 1 & -2 \\ -3 & 1 & 0 \end{bmatrix}$$

- **Parameters give a column space.** Put the coefficients of each parameter into one column:

$$\left\{ \begin{bmatrix} 2s - t \\ s + 3t \\ 4t \end{bmatrix} : s, t \text{ real} \right\} = \left\{ s \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix} \right\} = \operatorname{Col} \begin{bmatrix} 2 & -1 \\ 1 & 3 \\ 0 & 4 \end{bmatrix}.$$

$\operatorname{Nul} A$	$\operatorname{Col} A$
a subspace of $\mathbb{R}^n$ (inputs)	a subspace of $\mathbb{R}^m$ (outputs)
defined implicitly: the condition $A\mathbf{x} = \mathbf{0}$	defined explicitly: combinations of the columns
finding members takes row reduction	the columns are members; no work
$\mathbf{v} \in \operatorname{Nul} A$? Compute $A\mathbf{v}$.	$\mathbf{v} \in \operatorname{Col} A$? Row reduce $[\,A \;\; \mathbf{v}\,]$.
$\operatorname{Nul} A = \{\mathbf{0}\} \iff A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\iff \mathbf{x} \mapsto A\mathbf{x}$ is one-to-one	$\operatorname{Col} A = \mathbb{R}^m \iff A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$ $\iff \mathbf{x} \mapsto A\mathbf{x}$ is onto $\mathbb{R}^m$

The question answered. By the row-vector rule,

$$A\begin{bmatrix}2\\1\\0\\0\\0\end{bmatrix} = \begin{bmatrix}2-2\\0\\4-4\end{bmatrix} = \mathbf{0},$$

so $(2,1,0,0,0)$ is in $\operatorname{Nul} A$ (it is the first spanning vector). For the column space, apply the same row operations to $[\,A \;\; \mathbf{v}\,]$, first with $\mathbf{v} = (1,2,4)$ and then with $\mathbf{v} = (1,2,3)$:

$$\left[\begin{array}{ccccc|c}1&-2&0&-1&3&1\\0&0&1&2&-2&2\\2&-4&1&0&4&4\end{array}\right] \to \left[\begin{array}{ccccc|c}1&-2&0&-1&3&1\\0&0&1&2&-2&2\\0&0&0&0&0&0\end{array}\right],$$

$$\left[\begin{array}{ccccc|c}1&-2&0&-1&3&1\\0&0&1&2&-2&2\\2&-4&1&0&4&3\end{array}\right] \to \left[\begin{array}{ccccc|c}1&-2&0&-1&3&1\\0&0&1&2&-2&2\\0&0&0&0&0&-1\end{array}\right].$$

So $(1,2,4)$ is in $\operatorname{Col} A$ (it is column 1 plus twice column 3), and $(1,2,3)$ is not (last row $0 = -1$). All solutions of $A\mathbf{x} = \mathbf{0}$ form $\operatorname{Nul} A$, the span of the three vectors above; all combinations of the columns form $\operatorname{Col} A$.

Linear transformations between vector spaces. $T\colon V \to W$ is **linear** if $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u})+T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$ for all $\mathbf{u}, \mathbf{v}$ in V and all scalars c. Its **kernel** is $\{\mathbf{u} \in V : T(\mathbf{u}) = \mathbf{0}\}$ and its **range** is $\{T(\mathbf{x}) : \mathbf{x} \in V\}$.

Theorem (Kernel and range (Lay 4.2))

The kernel of a linear $T\colon V \to W$ is a subspace of V, and its range is a subspace of W. For $T(\mathbf{x}) = A\mathbf{x}$ they are $\operatorname{Nul} A$ and $\operatorname{Col} A$.

Proof. 1. $T(\mathbf{0}) = T(0 \cdot \mathbf{0}) = 0\,T(\mathbf{0}) = \mathbf{0}$, so $\mathbf{0}$ is in the kernel and in the range.

2. Kernel: if $T(\mathbf{u}) = T(\mathbf{v}) = \mathbf{0}$, then $T(\mathbf{u}+\mathbf{v}) = \mathbf{0}+\mathbf{0} = \mathbf{0}$ and $T(c\mathbf{u}) = c\mathbf{0} = \mathbf{0}$.

3. Range: $T(\mathbf{x}) + T(\mathbf{y}) = T(\mathbf{x}+\mathbf{y})$ and $cT(\mathbf{x}) = T(c\mathbf{x})$ are again images.

4. For $T(\mathbf{x}) = A\mathbf{x}$ the kernel is $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\} = \operatorname{Nul} A$ and the range is $\{A\mathbf{x}\} = \operatorname{Col} A$.

□

Worked example: differentiation. $D\colon \mathbb{P}_3 \to \mathbb{P}_2$, $D(p) = p'$, is linear because $(p+q)' = p'+q'$ and $(cp)' = cp'$.

1. Kernel: for $p = a_0+a_1t+a_2t^2+a_3t^3$, $p' = a_1+2a_2t+3a_3t^2$ is the zero polynomial exactly when $a_1 = a_2 = a_3 = 0$. So the kernel is the constant polynomials.

2. Range: every $b_0 + b_1t + b_2t^2$ is a derivative, namely

$$\frac{d}{dt}\left(b_0t + \frac{b_1}{2}t^2 + \frac{b_2}{3}t^3\right) = b_0 + b_1t + b_2t^2,$$

so the range is all of $\mathbb{P}_2$.

In the same way, $y \mapsto y'' + \omega^2 y$ is linear from the twice differentiable functions to the functions, so the solution set of $y'' + \omega^2 y = 0$, its kernel, is a subspace.

Shortcuts

- **Spanning set for** $\operatorname{Nul} A$: one vector per free variable, from the parametric vector form; it is automatically independent. Reason: the paragraph above. Instance: the question.
- **Membership tests**: $\mathbf{v} \in \operatorname{Nul} A$? Compute $A\mathbf{v}$. $\mathbf{v} \in \operatorname{Col} A$? Row reduce $[\,A \;\; \mathbf{v}\,]$. Reason: the definitions and (W4). Instance: the question.
- **Recognize the space**: homogeneous equations give Nul; parameters give Col; either way it is a subspace. Instance: the two examples above.
- **Sizes**: for $m \times n$ A, $\operatorname{Nul} A \subseteq \mathbb{R}^n$ (number of columns) and $\operatorname{Col} A \subseteq \mathbb{R}^m$ (number of rows). Instance: for the question's 3×5 matrix, $\operatorname{Nul} A \subseteq \mathbb{R}^5$ and $\operatorname{Col} A \subseteq \mathbb{R}^3$.

Traps

- A vector with the wrong number of entries cannot be in the space: $(1, 2)$ is never in $\operatorname{Col} A$ for a matrix A with three rows, whatever its columns.
- The columns of the reduced echelon form do not span $\operatorname{Col} A$ in general. For the matrix below, the RREF has columns $(1, 0)$ and $(2, 0)$, spanning the x_1-axis, but $\operatorname{Col} A = \operatorname{Span}\{(1, 2)\}$.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

- $A\mathbf{x} = \mathbf{b}$ consistent for one $\mathbf{b}$ does not give $\operatorname{Col} A = \mathbb{R}^m$: for the question's matrix $(1, 2, 4)$ is in $\operatorname{Col} A$ but $(1, 2, 3)$ is not.

§3 Homework this week

HW4, Sections 3.2, 3.3, 4.1, and HW5, Section 4.2.

Section 3.2.

- **#2, #3** (which row operation?). Read off the operation, then Theorem 3 (a factor pulled out of a row is (c); a replacement is (a)).
- **#5, #9** (determinant by reduction). Replacements and interchanges only; $\det = (-1)^r \cdot$(diagonal product).
- **#15, #19, #20** (from a 3×3 matrix with known determinant). Name the operation that produced each matrix and apply Theorem 3.
- **#23, #24** (invertible? independent?). Theorem 4; for vectors, put them in as columns.
- **#27, #29, #31** (true/false). Theorem 3(a); Theorem 4 with dependent columns; three interchanges give $(-1)^3$.

Section 3.3.

- **#3, #5** (Cramer's rule). One determinant $A_i(\mathbf{b})$ for each unknown.
- **#9** (Cramer's rule with s). $\det A$ as a polynomial in s; unique exactly when it is nonzero (worked example).
- **#11** (inverse by the adjugate). Nine cofactors, then transpose (worked example).
- **#21** (parallelogram from vertices). Move one vertex to $\mathbf{0}$; the two adjacent sides are the columns.
- **#27** (area of an image). Theorem 10: multiply by $|\det A|$.
- **#29, #30** (triangle areas). Half a parallelogram; the triangle shortcut.

Section 4.1.

- **#2, #3, #6, #7** (subspace?). Find the condition that fails: $\mathbf{0}$, sums, or multiples (traps of 4.1).
- **#10, #11** (sets given by parameters). Write the set as a span; Theorem 1.
- **#20** (functions on $[a, b]$). The zero function, and sums and multiples of continuous functions with $f(a) = f(b)$.

- **#23, #28, #32** (true/false). The zero vector of a function space is the zero function; V is a subspace of itself; the conditions must hold for all vectors.

Section 4.2.

- **#1** (is $\mathbf{w}$ in $\operatorname{Nul} A$?). Compute $A\mathbf{w}$.
- **#5** (spanning set for $\operatorname{Nul} A$). One vector per free variable.
- **#9, #13, #15** (write a set as Nul or Col). Homogeneous equations give a matrix whose null space it is; parameters give the columns.
- **#17, #24** (which $\mathbb{R}^k$; membership). Sizes shortcut; compute $A\mathbf{w}$, or row reduce $[\,A \quad \mathbf{w}\,]$.
- **#25, #31, #32, #36, #38** (true/false). Definitions; consistency for *every* $\mathbf{b}$; linearity of differentiation; $\operatorname{Row} A = \operatorname{Col} A^T$.
- **#40** (a second solution). If $A\mathbf{x} = \mathbf{b}$ then $A(5\mathbf{x}) = 5\mathbf{b}$.
- **#43, #46** (linear maps on polynomials and functions). Check the two rules; find the kernel from the defining condition, as in the differentiation example.

§4 Problem set

P1. Compute the determinant below by row reduction.

$$\det \begin{bmatrix} 2 & 4 & -2 & 6 \\ 1 & 3 & 0 & 2 \\ -1 & -1 & 4 & 1 \\ 3 & 8 & 1 & 5 \end{bmatrix}$$

P2. A and B are 3×3 with $\det A = 3$ and $\det B = -2$. Find $\det(AB)$, $\det(A^{-1})$, $\det(A^T)$, $\det(2A)$ and $\det(B^T A^{-1} B)$.

P3. For which h is the matrix below invertible?

$$\begin{bmatrix} 1 & 2 & h \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

P4. Use Cramer's rule to solve $5x_1 + 2x_2 = 1$, $3x_1 + x_2 = 4$.

P5. Find the area of the parallelogram with vertices $(-1, 0)$, $(0, 5)$, $(1, -4)$, $(2, 1)$. Then find the area of its image under the map below.

$$T(\mathbf{x}) = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \mathbf{x}$$

P6. Which are subspaces? (a) $\{(a, b, c) \in \mathbb{R}^3 : a + b = 0\}$. (b) $\{(a, b) \in \mathbb{R}^2 : b = a^2\}$. (c) $\{p \in \mathbb{P}_2 : p(0) = 1\}$. (d) $\{(s - 2t,\ s,\ 3t) : s, t \text{ real}\}$. (e) The set of 2×2 matrices A with $A^T = A$.

P7. For the matrix below, find a spanning set for $\operatorname{Nul} A$, and decide whether $\mathbf{b} = (3, 3, -4)$ is in $\operatorname{Col} A$.

$$A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$$

P8. For the matrix below, $\mathbf{u} = (3, -2, -1, 0)$ and $\mathbf{v} = (3, -1, 3)$: is $\mathbf{u}$ in $\operatorname{Nul} A$? Is $\mathbf{v}$ in $\operatorname{Col} A$? Could $\mathbf{u}$ be in $\operatorname{Col} A$?

$$A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$$